DE.AUTO

CHAPTER 9

BRAKES & SAFETY

THE WORLD OF CAR MODIFICATIONS MADE SIMPLE

Cars explained in plain, friendly English

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POWERED BY DE.AUTO

DE.AUTO — Because knowledge is power, and cars are passion.

CAR MODIFICATIONS GUIDE
MUST-KNOW TERMS

Car modifications, explained

Chapter Nine: Brakes and Safety

“GO FAST ONLY IF YOU CAN STOP FASTER.”

— DE.AUTO

Important notice

This guide is **entertainment only**. It is not educational or scientific content, and it is not a how-to manual. DE.AUTO is not an accredited scientific, engineering or technical authority. Everything here is simplified for fun and general interest, and it can differ from one car to another.

You must **never** try any modification, drive at high speed, or do any kind of stunt or dangerous driving on public roads. Safety and respecting traffic laws are everyone's own responsibility.

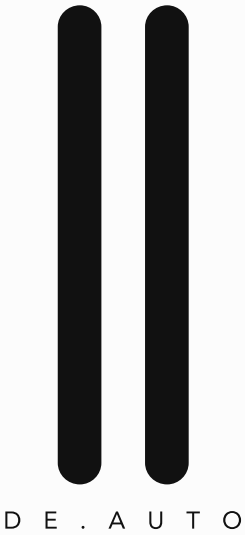
The information here is **general and approximate**, and much of it describes only the typical case. Laws, regulations and specifications differ by country and model, so always check the owner's manual and the official authorities.

Vehicle modifications are subject to local laws. Any unauthorized modification can lead to legal liability, fines or having your vehicle impounded.

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For any decision about your car, go to certified professionals and the manufacturer's manual, and only try something if it complies with official regulations and legal requirements. 🚀



DE.AUTO is more than a name. It's an identity that brings the passion for cars together with knowledge and progress.

At DE.AUTO we make fresh content for car lovers, mixing fun and know-how from the world of modifications and high performance, in a simple style that's easy to follow. Our goal is to help you understand your own car better and make smarter calls on modifications and maintenance.

We walk you through the key terms and parts of the modification world, so you can chat with the enthusiasts and follow every detail without feeling like an outsider.

Whether you're a beginner or a pro, you'll find terms, basics and fun facts here that help you get into cars in an organized, confident way.

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READY?

You stamp on the brakes with everything you've got... and the car still doesn't stop. What was missing? 😳

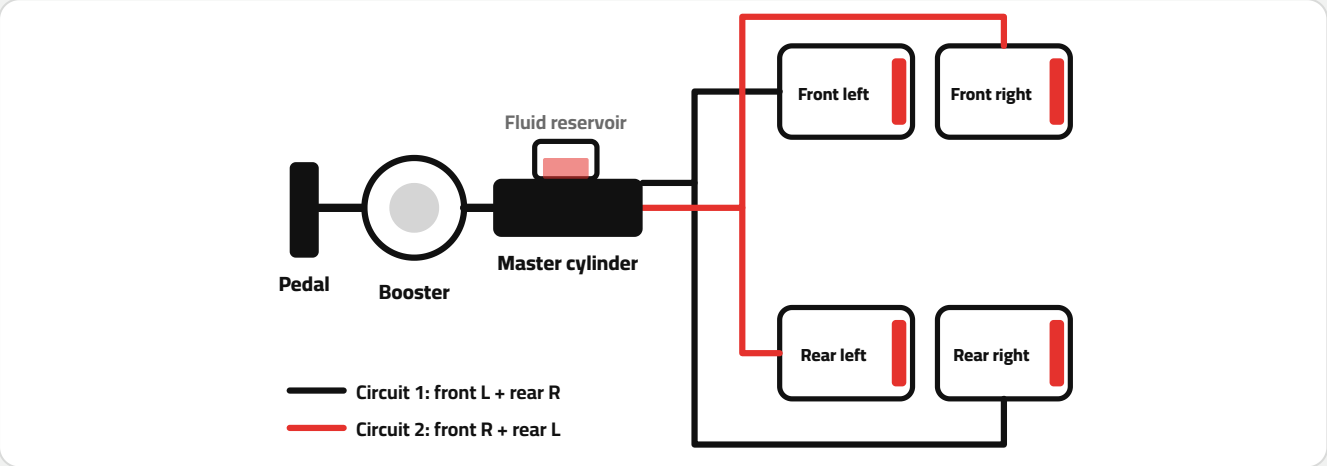
By the end of this chapter you'll know:

- ✓ How foot pressure becomes fluid pressure that stops a ton of steel.
- ✓ Discs, pads and calipers: what really helps and what is only looks.
- ✓ Brake fluid: why it ages, and the DOT types on the label.
- ✓ Why tires stop the car before any big brake kit does.
- ✓ What the pedal does when the ABS kicks in, and why you never pump it.
- ✓ How ESC catches a skid before you feel it, and which systems modifications can affect.



01 From your foot to the wheel: the hydraulic circuit 💧

In Chapter 1 you learned that fluid transmits pressure. Now we go inside the system and look at every part between your foot and the tire:



Simplified illustration: diagonal split of the two circuits. Some cars split them front/rear. The circuit shown has no ABS, for simplicity

1. Your foot presses the pedal. The **booster** multiplies your force, usually with vacuum from the engine, and in other cars with a pump or an electric motor.
2. The pedal pushes two pistons inside the **master cylinder**, so the fluid pressure rises.
3. The fluid travels through the **brake lines** (metal pipes, with flexible hoses at the wheel).
4. At each wheel, the caliper piston squeezes the pads against the disc, and the friction slows the wheel.

🛡️ **Built-in safety:** The master cylinder has **two independent circuits**. If a line is cut in one, the other still brakes, though with less force. That's so you don't lose all your braking at once.

×4

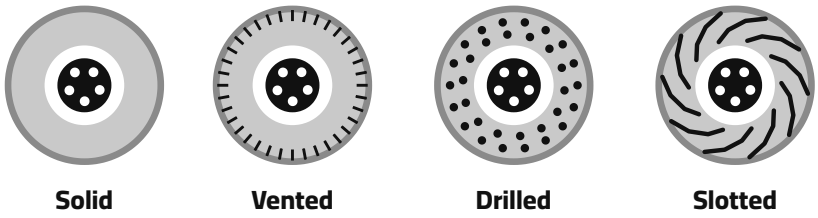
Double your speed and the car's kinetic energy becomes four times bigger, so the brakes (and the distance) need roughly four times as much to stop it. 100 km/h isn't "double" 50, it's **four times** the energy. 😮

🎯 Challenge 1 **If your speed goes from 50 to 100 km/h, roughly what happens to your braking distance?**

A) It doubles | B) It quadruples | C) It doesn't change

02 Brake discs: four shapes, and each has a story 💿

The brake disc (also called the rotor) is what carries the heat away. Its weight, material and design decide how much it can take. The shapes you will meet:



Simplified illustration of the shapes (on the vented disc you can see the inner vanes at the rim)

Type	The appeal	The truth you need to know
Solid	A single disc with no gap. The simplest, cheapest and lightest; suits the rear axle and light cars.	Cools more slowly, so it can't take repeated hard braking.
Vented	Two discs with vanes between them that pull air through and cool from the inside. The usual choice on the front axle.	Heavier and more expensive than solid.
Drilled	Holes that let water, gas and dust escape from the friction surface, and it looks great.	The holes are weak points: small cracks can appear around them with heat and repetition, especially in cheap quality and hard use.
Slotted	Slots that scrape and clean the pad surface and let gas out. Less prone to cracking than drilled.	Can add friction noise and wear the pads faster.

💡 **Did you know?** On expensive cars you find a "floating-bell disc" (aluminium centre with an iron friction ring) to save weight and let the ring expand freely with heat. On very high-performance and very expensive cars you find carbon-ceramic discs, which handle extreme heat, cost a lot, and suit a specific kind of use.

❌ "Drilled or slotted discs always shorten the distance." Not accurate: the distance is decided by the tires first, then by how much friction and heat the brakes can handle. The shape improves venting or looks, not necessarily stopping.

03 Brake pads: friction has a temper 🧱

The brake pad is the part that rubs and wears itself away to stop you. Every compound (mix) carries a **trade-off**: no pad wins at everything.

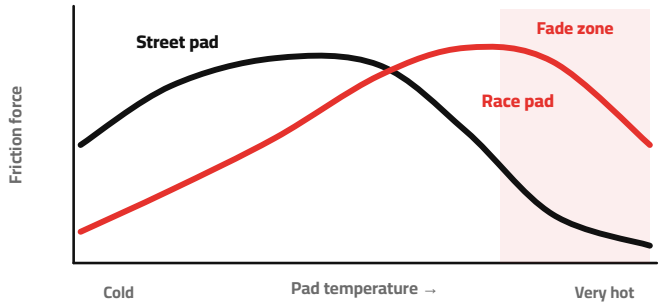
Organic
Quiet and gentle on discs, but may wear faster. For light daily street use.

Ceramic
Quiet, clean, low dust and usually long-lived. Daily street use; the bite varies by maker.

Semi-metallic
Handles more heat with a stronger bite, but makes more dust, noise and disc wear.

Race pads
Need heat to work. Cold, they bite weakly, and on the street can be noisy and dusty.

"Dust-free pads"
A marketing term. Every pad leaves dust; some just leave less, and finer.



Idea only (no numbers): a street pad bites when cold but fades when hot; a race pad is the opposite

Cold bite: the braking force the moment you press, with the pad still cold. It's the one that matters day to day.

Fade: friction drops when the pad goes past its working temperature, so the same pedal force slows the car less.

🎯 Challenge 2 **You fit race pads on a street car and brake on a cold morning. What happens?**

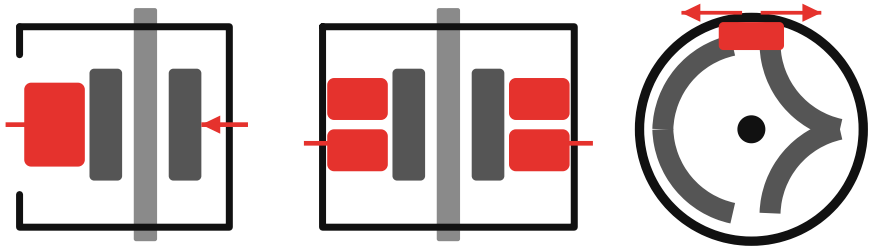
A) Strong bite from the very first moment

B) Weaker bite, because the pads haven't warmed up

C) No difference

✅ **DE.AUTO tip:** Choose the pad for how you really drive, and change pads for the whole axle together, never one side.

04 Calipers, drums and the handbrake 🔧



Floating
piston on one side

Fixed
pistons on both sides

Drum
shoes push outward

Caliper: floating or fixed?

Floating: Simpler, lighter and cheaper, and the common choice on most cars. One piston (or two) on one side, and the body slides on pins to press the other pad.

Fixed: A rigid body with pistons on both sides. Stiffer, with a more precise pedal feel, and usually found on high-performance and racing cars.

Number of pistons (four, six, more): the idea is that a longer pad needs more pistons so the pressure spreads evenly across it. The piston count alone doesn't shorten the stopping distance; the whole design and the tires are what count.

Drums: old, but not dead 🥁

Drum brake: two shoes push against the inside wall of the drum. Cheaper and tougher against dust, and the handbrake combines with them easily, which is why you often see them at the rear of economy cars. Their downside is that they shed heat more slowly than a disc, so they suffer more under repeated hard braking.

The handbrake: the parking brake 🅿️

Manual (cable)

A cable pulls the drum shoes or pushes the rear caliper piston. A simple mechanical cord.

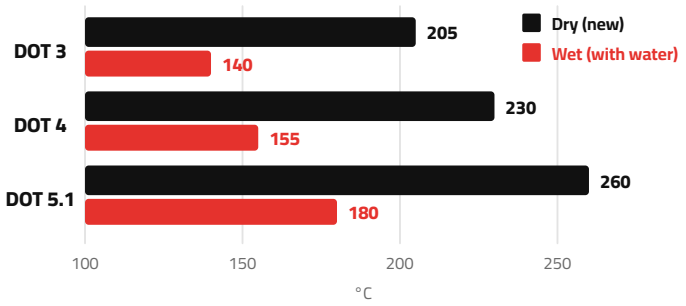
Electric (EPB)

A small button and a motor that clamps the pad. Comes with extras like auto-hold in traffic and on hills.

❌ On an electric handbrake, some cars need a "service mode" with a scan tool to change the rear pads. Pushing the piston back without it damages the motor. This is a job for specialists, not an experiment.

05 Fluid and lines: the brakes' lifeblood

Brake fluid must **not compress** and must not boil easily. But it has a flaw: it slowly absorbs moisture from the air, even inside a sealed system. And water boils early, so the fluid's whole boiling point drops.



Minimum approved boiling point for each type (approximate). Check your car's specification

140 °C That's the minimum wet boiling point of DOT 3 fluid after it absorbs moisture, versus 205 °C when new. Once fluid boils it makes vapour, and vapour compresses (fluid doesn't), so the pedal sinks to the floor. 😬

DOT 3 / 4 / 5.1: Glycol-based, and they differ in boiling point. Stick to what the manufacturer specifies.

DOT 5 is not DOT 5.1: The first is silicone and doesn't mix with glycol; the second is glycol. One digit, big difference.

Changing it: Periodically, per the manufacturer, and don't judge by its colour, because fluid that looks clean can still hold moisture.

Lines: rubber or braided?

Rubber hose
The standard on most cars. Cracks and swells with age and stretches slightly under pressure.

Braided (stainless)
A braided metal sheath reduces stretch so the pedal feels firmer. Must be approved and road-legal.

🫧 **The idea of bleeding:** Air compresses and fluid doesn't. If air gets into the lines the pedal feels spongy, which is why the bubbles are expelled (bled) with a proper procedure. This is work on a safety system, so leave it to a qualified shop.

06 Fade, cooling and balance 🔥

Brakes turn motion into heat. If they can't get rid of it at the same rate, they start to tire. It comes in three forms:

Pad fade
Friction weakens because the pad has gone past its working temperature. The pedal is firm but the car doesn't slow with the same force.

Fluid fade
The fluid boils and forms vapour. The pedal goes down further and feels "soft" or sinks.

Mechanical fade
Heat expands the parts so the shape changes, and in drums they expand and move away from the shoes.

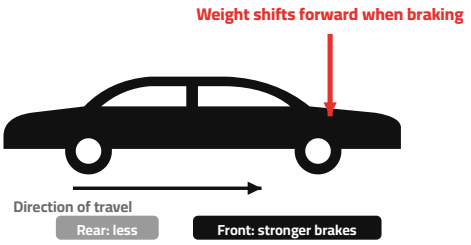
How are brakes cooled? 💨

Air: Open wheels and air ducts in the bumper (on performance cars) direct a stream of air at the disc.

After hard use on track: Do a cool-down lap before parking, and don't hold the brakes on while stopped with a very hot disc, because the pad leaves a deposit on its surface and causes vibration.

Brake balance: why is the front stronger? ⚖️

When you brake, weight shifts forward and the front tires get pressed harder, so they can handle stronger braking. That is why the front brakes are bigger and stronger than the rear on most cars.



Simplified diagram of weight transfer

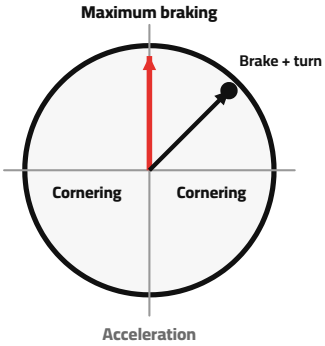
If the rear is too strong: The rear wheels lock before the front, the back of the car slides out and spins. That is more dangerous than the opposite.

If the front is too strong: The front locks first and you lose steering (the car carries on in a straight line).

On road cars the factory sets the balance, and EBD (coming up shortly) refines it.

07 The big brake kit: what does it really improve? 🛞

The biggest myth on this subject: “bigger disc = shorter stop.” Let’s be frank: **the tires are what stop the car.** The brakes turn your motion into heat, but it is the tire that passes the stopping force to the tarmac. Four small patches decide everything.



The grip circle: the tire has one “budget”. What goes into braking is taken away from cornering

A big kit improves:

Heat capacity: A bigger, heavier disc soaks up more heat and cools it over a bigger area. Reduces fade in repeated hard braking.

Consistent feel: A firmer pedal that you can control more precisely.

Looks: And that’s true, we won’t deny it. 😄

And what it doesn’t necessarily improve:

Stopping distance: If your tires are already at their maximum grip, a bigger disc doesn’t add any. Better tires make a bigger difference.

Compatibility: It has to suit the wheel, the master cylinder and the front/rear balance, otherwise the car’s behaviour changes.

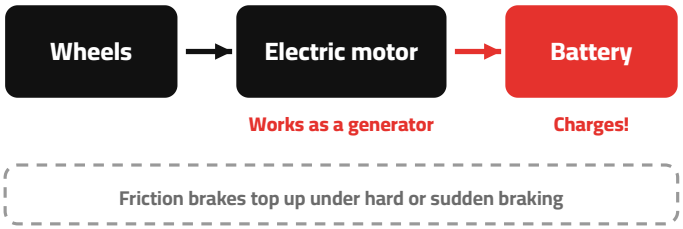
Legality: Approved parts, professional fitting, and in line with local regulations and insurance.

🎯 Challenge 3 **You have old, hardened tires. Which upgrade makes the biggest difference in an emergency stop?**

A) Bigger discs | B) Suitable new tires | C) A caliper with more pistons

08 Electric braking and regeneration ⚡

In hybrids and EVs, the electric motor runs in reverse when you lift off or brake gently: instead of using energy, it **generates** electricity and charges the battery. It's called regenerative braking.



The simplified idea

Blended braking: The car mixes regeneration and the regular brakes depending on how hard you press, and most of the time the pedal feels normal.

Limits: A full or cold battery reduces regeneration, so the regular brakes have to fill in.

Watch out: Using the regular brakes less exposes the discs to more rust, so ask your technician about a periodic inspection.

Your car talks to you: signs and sounds 👂

Sign	Likely cause	What to do
Sharp squeal when braking	The pad wear indicator (a metal tab) touches the disc	Time to inspect and replace the pads soon
Metal grinding	The pad is gone and metal rubs on the disc	Stop soon and inspect right away. The disc is being damaged
Pulling to one side	A stuck caliper or a pressure difference between the two sides	Urgent inspection
Vibration in the pedal or steering wheel	Uneven disc thickness or deposits	Inspection and resurfacing or replacement
Spongy or sinking pedal	Air, moisture, a leak or boiling fluid	Don't keep driving without an inspection

❌ "I'll wait until it makes a noise": some pads wear down without any squeal. Check them according to the service schedule, and never ignore the brake or ABS warning light.

09 ABS and its siblings: the brain behind the pedal 🧠

ABS: preventing wheel lock

We promised you the details back in Chapter 1. When a wheel locks and skids, steering stops working and the car carries on in whatever direction it was going, even if you turn the wheel. The ABS (Anti-lock Braking System) detects a wheel that's about to lock through the wheel-speed sensors, eases its pressure off for a moment, then restores it **several times a second**, so the wheel keeps turning right at the edge of grip and you keep your steering.

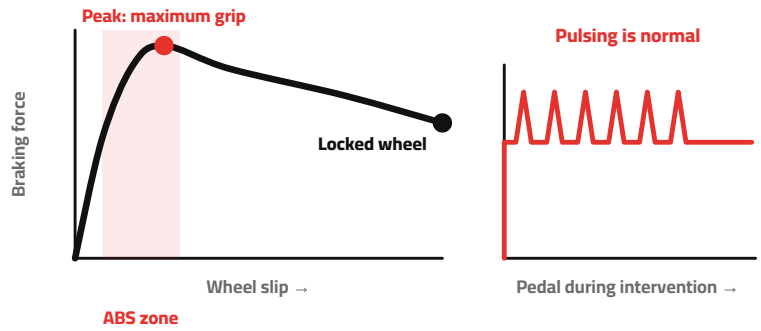


Illustration of the idea: peak grip comes just before the wheel locks, and the system keeps you around it. The pedal pulses while ABS intervenes

✅ **The golden rule: If you feel the pulsing: don't pump and don't lift.** Press the pedal hard and keep your foot planted, and steer where you want to go. That pulsing means the system is working for you. In old cars without ABS pumping used to help; now it hurts.

EBD (electronic brake-force distribution): shares the braking pressure between front and rear according to load and grip.

BA (brake assist): picks up how fast you press, and if it's an emergency stop it raises the pressure to the maximum.

TCS (traction control): detects a wheel spinning under acceleration, so it brakes it and/or cuts engine torque.

Reminder: ABS doesn't cancel physics, and on gravel or light snow the distance can get longer.

🎯 Challenge 4 **During hard braking you feel the pedal pulsing under your foot. What do you do?**

A) Pump the pedal quickly | B) Press hard, keep my foot planted and steer | C) Lift my foot

10 ESC: smarter than your reflexes 🌀

Electronic stability control (ESC), sold under different brand names by different makers, is one of the most important modern safety systems and helps reduce skids and loss of control. It works like this:

1. **The sensors read** the steering angle (what you want), each wheel's speed, and what the car is really doing: its yaw (rotation about its own axis) and its sideways acceleration.
2. **The computer compares** what you want with what the car is actually doing.
3. If there's a difference (a skid), it may brake **one or more wheels** precisely and cut engine torque if needed, bringing the car back onto its line.

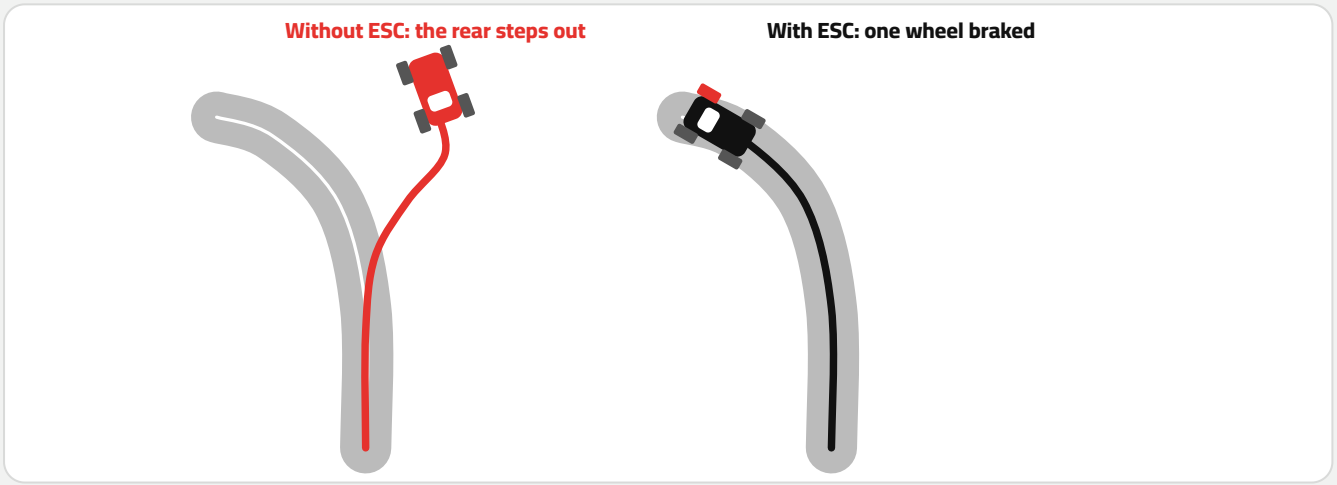


Illustration from above: the red wheel is the one being braked. Example of a rear-end slide (oversteer)

Understeer

A common example: the car doesn't turn enough and carries on straight. The system brakes an inside rear wheel and eases off the torque, which helps it rotate.

Oversteer

A common example: the rear slides outward. The system brakes the outside front wheel, which corrects the rotation.

❌ Switching the ESC off "for fun" on public roads is a risky decision. If its light is flashing, it's correcting for you right now, so slow down.

🎯 Challenge 5 **If the ESC light flashes while you're driving, what does it mean?**

A) A brake fault | B) The system is correcting a skid right now | C) A weak battery

11 Airbags, seat belts and driver-assist systems 🛡️

Brakes prevent the crash, and these systems reduce the damage if one happens. They all work as a team:

20–50 MS

Roughly the time (in milliseconds) an airbag needs to inflate after the system senses an impact. Faster than the blink of an eye! 👁️

Airbags (SRS)
A gas generator inflates a bag in a fraction of a second. Works with the belt, not instead of it. Sit a sensible distance from the wheel.

Belt pretensioner
Tightens the belt at impact to hold you in your seat, and the load limiter eases it slightly to reduce chest pressure.

❌ Don't change the steering wheel, tamper with the seats, or cover the side airbags with unapproved accessories: the airbag could fail or deploy wrongly. The SRS light means get it checked, and after any serious impact have the system inspected even if the airbag didn't deploy.

Driver-assist systems (ADAS) 📡

Automatic emergency braking (AEB): radar or a camera spots a collision risk and warns you or brakes.

Lane keeping (LKA): a camera reads the lane markings and warns you or gently corrects the steering.

Blind-spot monitoring and adaptive cruise (BSM / ACC): warns you about traffic beside you and keeps your distance from the car ahead.

So what does this have to do with modifying? 🔧

These systems are **calibrated** to your car's original dimensions, and any change can throw off what they read:

Wheel and tire size changes the wheel diameter, which skews the wheel-speed readings that the ABS and ESC rely on.

Suspension (lowering or raising): changes the angle of the radar and camera and how the car behaves.

Glass and bumpers: After replacement or repair, the camera and radar need recalibration.

Engine tuning (ECU Tune): can change the engine torque that the TCS and ESC have to deal with. Details are in Chapter 12. After any of these modifications, ask a specialist about calibration.

12 Myths and garage talk 🗣️

✕ MYTH	✓ FACT
"ABS means I can't crash."	ABS keeps your steering during braking, but it doesn't raise grip above the limits of the tire and road. Distance is still distance, and attention is still a must.
✕ MYTH	✓ FACT
"A drilled disc always stops shorter."	The tires, the pad and the heat decide. Holes change venting and looks, and can weaken the disc if its quality is poor.
✕ MYTH	✓ FACT
"Any brake fluid works in any car."	You must use the specified fluid. DOT 5 (silicone) is not DOT 5.1 (glycol), and mixing the wrong ones damages the system.

Garage talk vs. the proper name 🛠️

Gulf garage talk	Proper name	What it means
Teel / fahmat	Brake pads	The part that rubs against the disc
Hoob / disk	Brake disc (rotor)	The metal disc that turns with the wheel
Jilda	Brake shoe	The drum-brake equivalent of a pad
Fakk / caliper	Brake caliper	Holds the piston and the pads
Tarumba	Master cylinder	Turns pedal force into fluid pressure
Servo	Brake booster	Multiplies your foot's force
Hand brake	Parking brake	Manual or electric
Thabat ("stability")	Stability control (ESC)	Corrects skids
Tafreegh ("emptying")	Bleeding the brakes	Expelling the bubbles from the fluid

> 💡 Garage names differ from region to region, so if you hear a name you don't recognise, ask the technician to show you which part they mean.

Wrap-up 📌

Now you know the brakes aren't "one part", they're a whole chain: pedal, booster, master cylinder, lines, caliper, disc and pads, with an electronic brain on top: ABS, EBD, ESC. You also learned that the tires are what stop the car first, that brake fluid wears out, that fade is real, and that any change to wheels or suspension can affect the safety systems. 🚀

🎯 Challenge answers:

1) Roughly four times as much.

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2) Weaker bite, because the pads haven't warmed up.

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3) Suitable new tires.

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4) Press hard, keep my foot planted and steer, and don't pump.

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5) The system is correcting a skid right now, so slow down.

➡️ SOON We've said it many times: the tires! In **Chapter 10: Tires and Wheels** we open up the whole subject: tire compounds, air pressure, wheels, and even the wheel angles (Camber / Toe / Caster) and spacers. Four small patches, and the whole story lives in them. 🔥